

Module 05: Disk Scheduling — LOOK and C-LOOK

1. LOOK vs C-LOOK Summary

Algorithm	Boundary Behavior	Total Movement (Sample Queue)	Key Advantage
SCAN	Always goes to extreme 0 or 199	236 cylinders	Standard elevator
LOOK	Reverses at last request	208 cylinders	Eliminates empty boundary travel
C-SCAN	Fly-backs between 199 and 0	382 cylinders	Uniform wait time
C-LOOK	Fly-backs between max & min request	322 cylinders	Uniform wait + 60 tracks saved

Disk Scheduling: LOOK vs C-LOOK (Optimized Boundary Traversal)

Queue: 98, 183, 37, 122, 14, 124, 65, 67 | Head = 53 | Direction: Towards Lower Cylinders

LOOK Algorithm

• Optimization: Arm disk ke extreme end (0) tak NAHI jata! Sirf aakhiri pending request (14) tak jata hai, fir reverse hota hai.

Path: 53 → 37 → 14 → 65 → 67 → 98 → 122 → 124 → 183

Calculations:

- Downward to lowest request (14): $(53 - 14) = 39$ tracks
- Reverse from 14 up to 183: $(183 - 14) = 169$ tracks

Total Movement = 39 + 169 = 208 Cylinders

Total Head Movement = 208 Cylinders

• SCAN (236) se 28 cylinders kam travel kiya!

• Unnecessary boundary track travel (0 to 14) eliminated.

C-LOOK (Circular LOOK)

• Optimization: 199 tak jane ke bajaye 183 par stop hota hai, aur 0 par jane ke bajaye directly 14 par jump karta hai!

Path (Moving Up): 53 → ... → 183 → 14 → 37

Calculations (Upwards direction):

- 53 to max request (183): $(183 - 53) = 130$ tracks
- Jump from 183 to min request (14): $(183 - 14) = 169$ tracks
- 14 to 37: $(37 - 14) = 23$ tracks

Total Movement = 130 + 169 + 23 = 322 Cylinders

• C-SCAN (382) se 60 cylinders kam travel kiya!

• Industry preferred practical circular disk scheduler.

2. Solved Numerical Output

- LOOK (downward): $(53 - 14) + (183 - 14) = 208$ Cylinders.
- C-LOOK (upward): $(183 - 53) + (183 - 14) + (37 - 14) = 322$ Cylinders.